Proposing a New Feature Engineering Technique for the Detection of Schizophrenia using EEG Signals

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***Abstract*—In this paper, a new feature engineering technique for electroencephalogram (EEG) signals is proposed to detect schizophrenia disorder. The feature engineering technique is based on the variation of EEG signals over time. The proposed framework has five steps. In the first step, the EEG signals are denoised using multi-scale Principal Component Analysis. Then, each channel of the EEG signals is used to extract 32 features. Next, feature concatenation is used to merge these extracted features. The nearest neighbor component analysis is used to select the best group of features. Finally, a feed-forward neural network is used for classification. The proposed framework is tested using 10-fold cross-validation to avoid bias in the results. The results show that the proposed framework, using the new feature engineering technique, classifies the EEG signals perfectly by achieving a classification accuracy of 99.6%. The proposed framework can be used by doctors in hospital offices for schizophrenia detection.**

I. Introdution

Schizophrenia is a chronic, disabling psychiatric disorder that severely incapacitates an individual's capacity for rational thinking, mental health, decision-making, and interpersonal functioning. It is estimated that 20 million individuals worldwide are affected, which is a major public health issue. Proper and prompt diagnosis of schizophrenia is still the greatest challenge to doctors in spite of the fact that treatment modalities exist that are capable of mitigating the symptoms of the disorder. A major challenge is that the diagnosis all too frequently depends on patient-reported symptoms and signs, and subjective clinical opinion, which are inherently unstable and prone to error.The non-invasive recordings of the electrical activity of the brain in the form of electroencephalogram (EEG) signals have become a promising neuropsychological instrument for the evaluation of brain function. They have several strengths, such as low cost, good temporal resolution, and simplicity of use in both hospital and outpatient settings. Since EEG signals can demonstrate brain pathology that is not apparent on routine clinical evaluation, they are an appealing option for developing diagnostic tests.

A. Problem Statement

The previous models for detecting schizophrenia based on EEG signals typically rely on Convolutional Neural Networks (CNN), which require substantial computational resources. This is one of the main disadvantages of these models (1). In this proposal, we introduce a new feature engineering technique for detecting schizophrenia, which significantly reduces the computational requirements compared to deep learning-based models (1).

B. Previous works

A chronic and complex neuropsychiatric illness, schizophrenia drastically incapacitates a person's ability to think, feel, and interact with others. It is usually accompanied by comorbid conditions of anxiety, depression, and sociopsychological issues (2). Three domains of symptoms of illusions, positive and negative emotional disturbances, and cognitive impairment are what afflicted persons normally present with. The derailment of cognition and emotions precedes psychotic symptoms in the two phases through which the illness takes its course (1).Since visual and auditory stimuli make up a considerable portion of human perception, schizophrenia's faulty sensory processing causes serious problems with reality interpretation (3). Vital domains including memory, communication, social interactions, and everyday conduct are also impacted by the condition, which frequently leads to social isolation and a lower quality of life. Around 20 million people worldwide suffer from schizophrenia, which can cause gradual brain damage if left untreated, according to the World Health Organization (4). Early symptom onset frequently interferes with social, professional, and educational growth.

TABLE I: A summary of previous works on schizophrenia detection.

Ref	Method	Classifier	ACC (%)
(5)	Three nonlinear features in the time domain	Linear Discriminant Analysis	87.50
(6)	Fast Fourier Transform, spectral power of rhythms	ROC	62.20
(7)	S-transforms, time-frequency features	Support Vector Machine	93.90
(8)	Statistical features of the frequency domain	Support Vector Machine	87.00
(9)	Combination of source-level and sensor-level features	Support Vector Machine	88.24
(10)	ϵ-complexity coefficients	Random Forest	83.60
(11)	Four entropies, Kolmogorov complexity, and fractal dimension	Support Vector Machine	88.50
(12)	Event-related potential features	Linear Discriminant Analysis	71.00
(13)	14 Nonlinear features	Support Vector Machine	92.91
(14)	Eleven-layered deep learning	Convolutional Neural Network	98.07
(4)	Short-time Fourier Transform, time-frequency features	Convolutional Neural Network	95.00
(15)	Empirical Mode Decomposition, five statistical features	Bagged Tree	89.59
(16)	Graph-based features	Random Forest	89.29
(1)	Empirical Mode Decomposition, entropy-based features	Support Vector Machine	93.00
(17)	Multivariate Iterative Filtering and Hjorth parameters	Support Vector Machine	98.90
(18)	Optimal wavelet filter bank and norm entropy	K-nearest Neighbor	99.21
(19)	Phase space dynamics of EEG signals, graphical features	K-nearest Neighbor	94.80

C. Challenges in Accurate Diagnosis

There is currently no clinical test that is widely recognized for the accurate diagnosis of schizophrenia. Mental health experts' clinical interviews and behavioral observations are the mainstays of diagnosis, but they are vulnerable to subjectivity and variability (20). Although cognitive tests are standardized by scales such as the Scale to Assess Unawareness of Mental Disorder (SUMD), these interviews are still qualitative and subject to discrepancies. Additionally, clinical differentiation of schizophrenia is also greatly impaired by the fact that it frequently overlies symptoms of other mental illness, such as deep depression and manic depressive illness (15). Although they are occasionally used, sophisticated imaging technology like Positron Emission Tomography (PET) and Magnetic Resonance Imaging (MRI) are not widely distributed and are expensive (21).

D. The Role of EEG in Neurodiagnostics

An efficient and affordable substitute for tracking brain activity in mental illnesses like schizophrenia is the electroencephalogram (EEG). EEG offers a direct measurement of neuronal electrical activity with a high temporal resolution, in contrast to MRI or PET. Due to its extensive use in clinics and hospitals, EEG offers a chance to create scalable and reasonably priced computer-aided diagnostic (CAD) systems for the diagnosis of schizophrenia.

E. Architecture of CAD Systems for Schizophrenia Detection

Three main steps are often included in EEG-based CAD systems: signal processing, feature extraction, and classification. Every step makes a substantial contribution to the diagnostic model's overall accuracy (ACC). Since feature extraction techniques identify the underlying patterns in EEG signals that could differentiate schizophrenia persons from healthy people, they are very crucial. Based on their processing domains, CAD procedures are generally divided into four groups: deep learning-based methods, time domain, frequency domain, and time–frequency domain.

1) Deep learning-based methods: EEG-based diagnosis has been transformed by deep learning, which learns complex patterns from high-dimensional data and automates feature extraction. Preprocessing time and human bias are decreased by deep learning models' ability to immediately extract complex spatiotemporal representations from raw EEG data, in contrast to typical machine learning models that rely on manually created features. These models have shown superior performance in classification tasks, although requiring significant processing resources:

a) Convolutional Neural Networks (CNNs): CNNs can learn spatial features from EEG signals better than most others. The networks employ convolutional layers to derive hierarchical features from the raw input. Excellent classification accuracy has been obtained in numerous research studies using CNNs for EEG diagnosis.The accuracy in classification for a CNN-based method that employed Short-Time Fourier Transform (STFT) features, for instance, was **95%** The viability of CNNs for EEG analysis is also supported by the **98.07%**

accuracy of deeper CNN models, such as the eleven-layer model (14).

b) Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM): Recurrent neural networks (RNNs) and their more sophisticated variant, Long Short-Term Memory (LSTM) networks, are very good at processing EEG signals since they are time-series data. The sequential character of EEG signals and temporal connections are captured by these networks, which are essential for the diagnosis of diseases like schizophrenia. In particular, LSTM networks are excellent at simulating long-range dependencies in the data, which aids in spotting patterns in the electrical activity of the brain that change over time.

c) Hybrid Deep Learning Models: To increase classification accuracy, hybrid models that draw on the advantages of various deep learning architectures have recently been proposed. The **CNN-LSTM** architecture is one such model that combines the temporal modeling capacity of LSTMs with the spatial feature extraction capabilities of CNNs. Due to their ability to concurrently capture the temporal and spatial dynamics of EEG data, these hybrid models have been demonstrated to provide even greater classification accuracies.

2) Time Domain Feature Extraction Techniques: Directly obtaining linear and nonlinear statistical information from unprocessed EEG signals is the core goal of time-domain analysis. These characteristics are essential for describing the intrinsic unpredictability and disorderly pattern of brain activity linked to schizophrenia. Numerous research have shown how successful time-domain approaches are in categorization tasks:

- Nonlinear chaotic features combined with Support Vector Machines (SVM) achieved a classification accuracy of **92.91%** (13).
- Graph-theoretic features resulted in an accuracy of **89.29%** (16).
- Event-related potential (ERP) features classified using Linear Discriminant Analysis (LDA) yielded an accuracy of **71%** (12).
- ϵ-complexity coefficients analyzed using Random Forest (RF) classifiers achieved **83.60%** accuracy (10).
- A combination of source- and sensor-level features with an SVM classifier led to an accuracy of **88.24%** (9).
- Autoregressive, band power, and fractal dimension features classified with LDA achieved **87.50%** accuracy (5).
- Entropy-based features such as Shannon entropy, Kolmogorov complexity, and spectral entropy classified with SVM resulted in **88.50%** accuracy (11).

3) Frequency Domain signal Processing: Using techniques like the Fast Fourier Transform (FFT), frequency-domain techniques convert EEG signals into spectral components for the purpose of examining power distribution within the alpha, beta, theta, and delta frequency bands, as these components can yield meaningful distinction between schizophrenics and normal. The following notable studies and findings shed light on how EEG signals in the frequency domain provide critical information:

- In an FFT study of EEG signals, a classification accuracy (ACC) of **62.60%** (6).
- In a study using simplified statistical feature extraction items from the power spectra of EEG signals, followed by SVM classification **87%** classification accuracy (8).
- Power spectral density (PSD) features in the theta and alpha frequency bands have been shown to differentiate EEG signals from the minds of healthy individuals and from those with schizophrenia, while other frequency bands show significant power distribution sensitivity during psychotic episodes.
- A hybrid method, consisting of frequency-domain analysis in conjunction with machine learning classifiers like Random Forest (RF), was shown to provide better classification accuracy and confirmed the independent nature of frequency-domain features when linked with advanced classifiers.

4) Time–Frequency Domain Approaches: Time–frequency domain techniques combine the strengths of both time-domain and frequency-domain analyses, offering a more comprehensive representation of EEG signals. These methods utilize transformations such as Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD), and the Stockwell Transform (S-transform) to perform multi-resolution signal analysis. These techniques are especially effective in capturing the non-stationary and time-varying nature of EEG signals, making them particularly useful for diagnosing schizophrenia. Some notable findings include:

- The Discrete Wavelet Transform (DWT), paired with Artificial Neural Networks (ANN), achieved **66%** classification accuracy (ACC) (22).
- The Stockwell Transform (S-transform) applied with autoregressive features resulted in a high **93.9%** classification accuracy (7).
- Empirical Mode Decomposition (EMD), combined with statistical features, achieved **89.59%** classification accuracy using a Bagged Tree (BT) classifier (15).
- Another study using entropy-based features extracted from EMD with SVM reported a classification accuracy of **93%** (1).
- An optimized wavelet filter bank combined with K-nearest neighbor (KNN) classification achieved a remarkable **99.21%** classification accuracy by capturing norm entropy from sub-bands (18).
- Adaptive multivariate iterative filtering, combined with Hjorth parameters and SVM, demonstrated **98.90%** classification accuracy (17).
- In addition to the common use of wavelets, time-frequency methods like the Short-Time Fourier Transform (STFT) and the Hilbert-Huang Transform (HHT) have been explored, with promising results in capturing both temporal and spectral features in schizophrenia detection.
- Hybrid time–frequency approaches that incorporate deep learning techniques, such as Convolutional Neural Networks (CNNs), have been increasingly employed, enhancing classification performance and providing end-to-end learning capabilities that reduce the need for manual feature extraction.

The time-domain methods rely on complex and manual feature

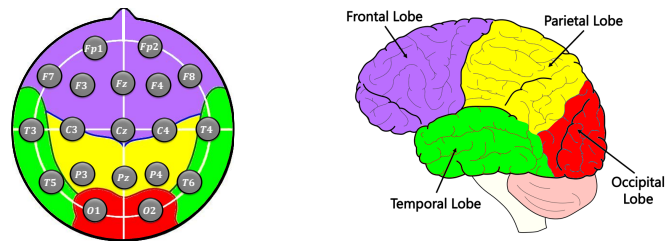
extraction, which is prone to error and noise, and also cannot fully quantify the non-stationary nature of EEG signals. These methods are also limited in quantifying behaviour unique to frequency. Spectral resoution will prevent frequency domain methods such as Fourier transform (FFT) from fully assessing low-frequency components; worse, they can reduce the signal to a single component which means they cannot assess time-varying components. While time-frequency domain analysis can be extended through the use of methods such as discrete wavelet transforms (DWT) and empirical mode decomposition (EMD) in the multiresolution fashion, those techniques rely on well-chosen features and are subject to overfitting and are computationally expensive. Deep learning-based methods rely on large labelled datasets and large computational resources, both of which are often limited in neurostimulation research. Additionally, deep learning based methods may generalise less well to new data because they are poorly interpretable, and because they can overfit for smaller labelled datasets.

II. Objectives of the Study

The aim of this research is to present a new feature engineering methodology to increase the accuracy and reliability of detecting schizophrenia. The new model is designed to extract features from EEG signals to be more discriminative features, offering improvements to diagnosis sensitivity and specificity. By utilizing various relevant signal processing methodologies and machine learning algorithms, our study aims to overcome the limits placed by traditional methods, which struggle to capture the complex and time-variant nature of EEG data. Our comparative analysis looks to establish the proposed model's capabilities against existing models, to see if it offers improved improvements of accurate and early detection of schizophrenia. The comparative analysis will focus on metrics of interest including but not limited to: classification accuracy, sensitivity (true positive rate, true negative rate), specificity (true positive rate, true negative rate) and computational efficiency. Overall, we aim to enhance the reliability and feasibility of detecting schizophrenia early with a method that enhances sensitively and cost-effectively utilize EEG signals.

III. The Used Database

A publicly available dataset is used, consisting of 14 files from healthy individuals and 14 files from individuals suffering from schizophrenia (8). EEG activity was recorded across 19 channels per the 10-20 system. The spatial major landmarks of the 19 channels including Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2 are displayed in Figure 1 depicting its 19 channels. The EEG signal has a sampling rate set at 250Hz and is 15 minutes long. Each EEG file has 19 by 225000 samples comprising a matrix where each column represents one of the 19 channels. Because EEG signals are known to be non-stationary, then we segment the non-stationary signals for each channel into 225 segments each composed from 1,000 samples. The process of signal segmentation by 1,000 samples occurs for all channels for each EEG file. In total 3,150 files of both people with schizophrenia and people without schizophrenia are available



(a) 10-20 standard (b) brain lobes

Fig. 1: Visualization of the location of electrodes on the skin in 10-20 standard and brain lobes.

for analysis. From this data we can develop and validate machine learning models aimed at detecting schizophrenia from EEG data. In addition, applying preprocessing techniques such as signal segmentation ensures that the data will be usable for feature extraction and classification tasks that contribute to the diagnostic benefits proposed by the model.

IV. Proposed Method

in this section we describe the step by step of the proposed method. Figure 2 shows the proposed framework for schizophrenia detection.

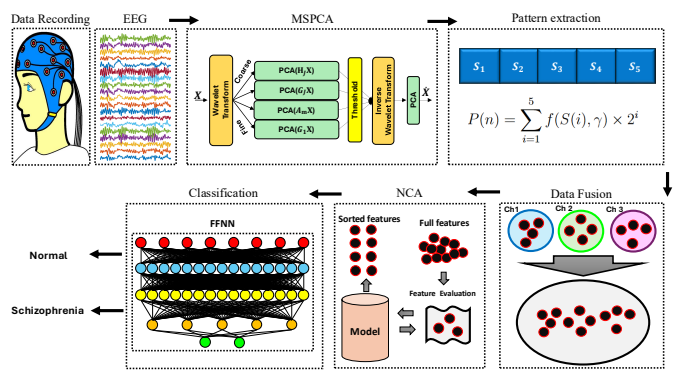


Fig. 2: Raw EEG Signal Visualization

A. Data Prepossessing

While the EEG signals contain near zero noise, (we will not address artifacts in this section, but we will refer to fraudulent signals as noise) in the pre-processing stage of neural-data it was deemed necessary to use multiscale PCA to clean the data and enhance the raw EEG signals - depending on artifact minimization. The EEG signal was created in 15-second sub-bands to help prevent ourselves from separating the different frequency components. When constructing the final EEG signal after the three sub-bands, we kept only those data pieces that were above a threshold and discarded those below - to further reduce artifact and noise data points that were unnecessarily corrupting the signal. The three sub-bands were then summed to generate a single EEG signal that was much clearer, while also retaining the essential quality of the signal required for performing all of the comparisons planned in

the first phase of the study. By employing this pre-processing method, we were also assured that the data extracted for the features and used for classification would minimize unwanted variation in the data, and therefore raise the expected performance of any models developed later on. This pre-processing approach also preserves the salient features of the signal and reduces the amount of data, while improving the reliability of any machine learning techniques employed. Figure shows the EEG signals in normal and schizophrenia groups.

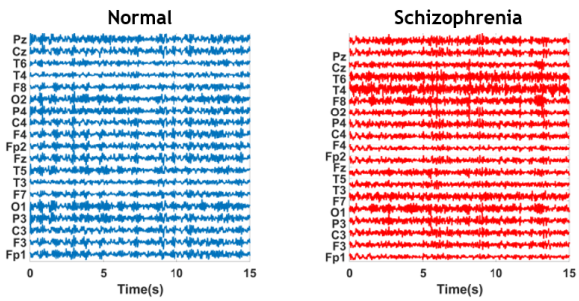


Fig. 3: Raw EEG Signal Visualization

B. Feature Extraction

In this research, we introduce a new feature extraction technique based on Local Binary Pattern (LBP). The objective is to characterize the exemplary patterns from the EEG signals, using the statistical properties of their local regions. First, we reshape the first 25 samples of the signal into a 5x5 matrix. After forming the matrix, we identify features that are important to classify later. For feature extraction, we first find the standard deviation of each column in the 5x5 matrix. Once the standard deviations are calculated for each column, we compare the two highest standard deviations and subtract one from the other. This step will enable us to isolate the regions in the signal that have variation, and those that are meaningful with respect to distinguishing the signal properties. Next, we calculate the standard deviation (total) of all 25 samples in the matrix. After understanding the total standard deviation, we can look at each column and compare the absolute value in the matrix to the total standard deviation. If the value in the column exceeds the total standard deviation, the binary number is 1; otherwise, the number is 0. Through the use of the binary number, representing the local variability in the signal employs a reliable feature to separate the normal and abnormal states of EEG. The Local Binary Pattern method is effective in encoding the spatial patterns within EEG data and has shown promise in distinguishing between different mental states, including schizophrenia. This approach is computationally efficient and does not rely on complex or computationally expensive procedures, making it suitable for real-time applications in medical diagnostics.

C. Data fusion

In this work, the data fusion is done by concatenating the features of each channel. Since the EEG signals have 19 channels and 32 features are extracted from each channel, a total of 608 features are extracted. Figure 5 shows the data fusion.

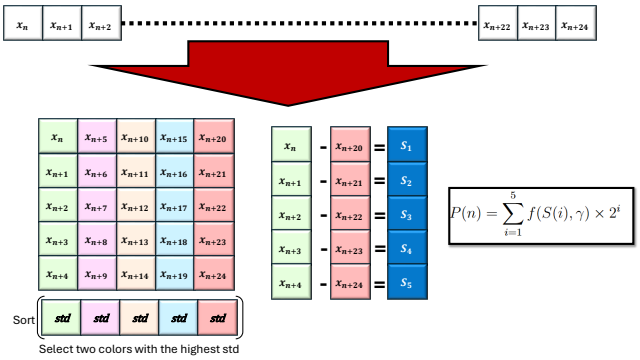


Fig. 4: Steps of the proposed feature extraction method.

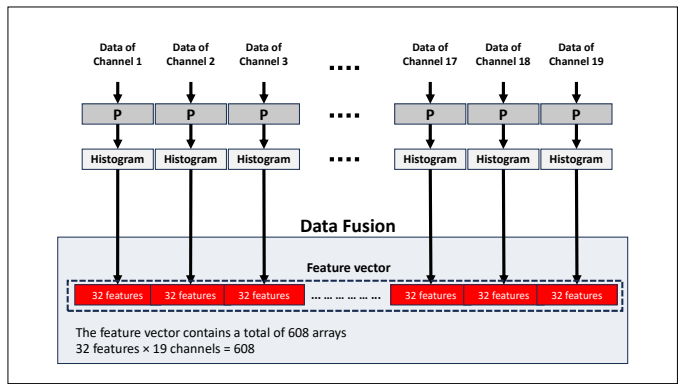


Fig. 5: Data fusion for extracted features from each channel.

D. Feature selection

The Neighborhood Component Analysis (NCA) is a supervised learning algorithm used for feature selection and dimensionality reduction. It learns a linear transformation of the input features that improves the performance of nearest neighbor classification. In this work, the NCA is used to sort the features from the best to the worst.

E. Classification

A Feedforward Neural Network (FFNN) is a simple type of artificial neural network used for classification. It takes input data, passes it through one or more hidden layers, and gives an output that tells us the class of the data. Each layer in the network has many nodes that process the information. The network learns by adjusting the connections between neurons during training. In this work, a FFNN architecture with one layer and two neurons is designed to classify the features in either normal or schizophrenia groups.

V. Results

The proposed method is evaluated in a ten fold cross validation (CV) to avoid bios in the results. we propose a new feature engineering technique for EEG signals to detect schizophrenia disorder. The feature engineering technique is based on the variation of EEG signals over time. In the first step of the proposed framework, the EEG signals are denoised using MSPCA. Then, each channel of the EEG signals is used to extract 32 features. Next, feature concatenation is used to merge these extracted features. The NCA analysis is used to select the best group of features. Finally, a FFNN is used for classification. The proposed framework is tested using 10-fold cross-validation to avoid bias in the results.

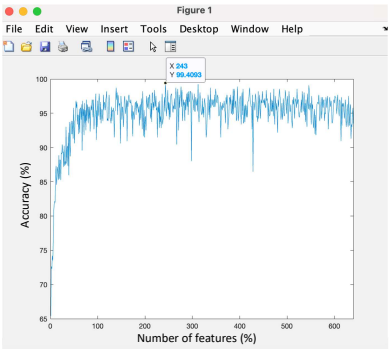


Fig. 6: Resulted accuracy for the selected features by NCA.

Figure 6 shows that the proposed framework, using the new feature engineering technique, classifies the EEG signals perfectly by achieving a classification accuracy of 99.6%. The highest classification accuracy is achieved using 243 features. We evaluated the p-value for these selected features using the NCA method. In other words, when the p-value is below 0.05, it means that the extracted features are statistically significant for discriminating between the normal and schizophrenia groups. This means that the null hypothesis is rejected, and the proposed feature engineering technique can properly distinguish between the normal and schizophrenia groups. Figure 7 shows the p-values for the selected features. It is clear that all the selected features by NCA are statistically significant to be used for classification. So, the proposed feature engineering technique to detect the schizophrenia rejects the null hypothesis and can be used for discrimination between the normal and schizophrenia EEG signals.

VI. Conclusion

In this study, a new framework has been proposed for automatic schizophrenia detection. By using the 10-fold CV technique, the FFNN classifier achieved the average classification ACC of 99.6%. Based on the accuracy we achieved for the proposed technique we can say that the feature engineering

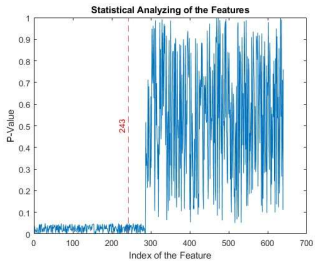


Fig. 7: P-values for the extracted features.

method can be used for schizophrenia detection by EEG signal. We check the p-value for the extracted features to reject the null hypothesis. As shown in the results, the p-value for the selected features by NCA was below than 0.05 which shows the rejection of the null hypothesis. This method can be used for several mental disorders detection such as seizure, depression, alcoholism and sleep apnea. The limitation of the proposed framework is the number of patients. In the future the performance of the proposed method will be tested by the database with more patients.

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